First Course Handout (FCH)

TA211
MANUFACTURING PROCESSES-I
2025-2026 Semester-I

Indian Institute of Technology, Kanpur

Department of Materials Science and Engineering

INSTRUCTOR	LAB IN-CHARGE
Prof. Shashank Shekhar **Office:** Western Labs-304 A **Ph:** 6528 / shashank@	**Mr. Rakesh Dixit** **Office:** DJC **Ph:** 7978 / kumarr@

Laboratory	**Monday, Wednesday, Friday: 14:00 - 17:00** **(As per assigned days of your section)** **Venue: TA211, Diamond Jubilee Complex** **(Engineering Metallurgy Laboratory)**

Course site	hello.iitk.ac.in

TA211: Introduction of Manufacturing Processes-I

Engineering Metallurgy Lab (Diamond jubilee Academic Complex)

*2025-2026, Semester-I, **Laboratory Schedule***

Time: 02:00 pm to 05:00 pm

Experiment Turns / Day & Section	1st E 1	2nd E 2	3rd E 3	4th E4	5th E5	6th Lab Exam + Drawing Submission	7th P 1	8th P 2	9th P 3	10th P 4	11th P 5	12th P 5	13th Project Evaluation
Monday	4/8	9/8[+]	11/8	18/8	25/8	1/9	8/9	6/10	13/10	27/10	1/11[+]	3/11	10/11
Wednesday	6/8	13/8	20/1	27/8	30/8[++]	3/9	10/9	24/9	8/10	15/10	22/10	29/10	12/11
Friday	1/8	8/8	22/8	23/8[*]	29/8	12/9	26/9	10/10	17/10[*]	24/10	31/10	7/11	14/11

Holidays, Breaks and Extra Working Day Schedule	Date	Saturday Extra Lab
Mid Semester Examination	Sep 16-22, 2025	**NA**
Mid Semester Recess (Dussehra)	Sep 27 – Oct 5, 2025	**NA**
End Semester Examination	Nov 17 – Nov 26, 2025	**NA**
[+]**DOAA Working Day (Monday Schedule)**	Aug 9, 2025 and Nov 1, 2025	**Monday Batch 2.00 pm to 5.00 pm**
[++]**Extra Lab – Wednesday (Wednesday Schedule)**	Aug 30, 2025	**Wednesday Batch 2.00 pm to 5.00 pm**
[*]**DOAA Working Day (Friday Schedule)**	Aug 23, 2025 and Nov 8, 2025	**Friday Batch 2.00 pm to 5.00 pm**

Lab In-Charge:
Mr. Rakesh Dixit
Contact:7978/ kumarr@

Course Instructor:
Prof. Shashank Shekhar
Contact:6528/ shashank@

Tutors, TAs & staff members contact:

Tutors:

Day	Tutor	Email	Phone
Monday	Prof. Shashank Shekhar	shashank@	6528
Wednesday	Prof. Kantesh Balani	kbalani@	7920
Friday	Prof. Heena Khenchandani	heenak@	2619

TAs:

Day	Before Midsem	After Midsem
Monday	Adam Musa (madam24@)	Raghav Mundra (rmundra20@)
	Suraj Kumar Chaudhary (surajkc24@)	Adam Musa (madam24@)
Wednesday	Adam Musa (madam24@)	Athul Raj R S (athulrajrs24@)
	Suraj Kumar Chaudhary (surajkc24@)	Suraj Kumar Chaudhary (surajkc24@)
Friday	Athul Raj R S (athulrajrs24@)	Athul Raj R S (athulrajrs24@iitk.ac.in)
	Raghav Mundra (rmundra20@)	Raghav Mundra (rmundra20@)

Staff members:

S No.	Staff Members	Email	Phone
1.	Mr. Rakesh Dixit	kumarr@iitk.ac.in	7978/7865
2.	Mr. Gaurav Mishra	mgaurav@iitk.ac.in	7978
3.	Mr. Bharat Raj Singh	brsingh@iitk.ac.in	7978
4.	Mr. Gyanendra Singh	gyans@iitk.ac.in	7978
5.	Mr. Anurag Prasad	anuragp@iitk.ac.in	7978/6277
6.	Mr. Surya Prakash Sonkar	spsonkar@iitk.ac.in	7978
7.	Mr. Rajdipta Samadder	rajdipta@iitk.ac.in	7978
8.	Mr. Pappu	--	7978

General Information

Grading policy:

- ***Laboratory (100%):***
  Weekly lab quiz and attendance (E1-E5): 15 %
  Weekly job (E1 – E4): 15 %
  Lab exam (LE6): 15 %,
  Weekly Project Design Progress (E2-E5): 10%
  Weekly Project Progress and attendance (P1 – P6): 15 %
  Project report (6th turn): 10 %
  Final Project Evaluation (PE) with weightage to attendance: 20 %
  Minimum 40% required for passing the course.
- There will be a total of 12 labs starting from 1st Aug (for Monday) and 6th Aug (for Wednesday). All the lab turns are mandatory. Only during medical emergencies, which must be approved by SUGC, absence can be accepted. In such cases, no make-up for attendance or quiz marks would be given. 13th Turn is reserved for Project Evaluation. Absence in 3 or more labs will result in deregistration OR Fail grade
- Contribution to the group project are mandatory components of the course.

Safety:

- Half pants, loosely hanging garments, and slippers are strictly prohibited for safety reasons. Students must come to the laboratory wearing (i) Trousers, (ii) Preferably cotton cloths (especially for welding/brazing) and (iii) Closed shoes.
- To avoid injury, the student must take the permission of the laboratory staff before handling any machine. Careless handling of machines may result in serious injury.
- Students must ensure that their work areas are clean and dry to avoid slipping.
- A leather apron and leather hand gloves will be issued to each student during Welding and Brazing exercises. Students not wearing the apron will not be permitted to work in the laboratory.
- At the end of each experiment, students must clear off all tools and materials from the work area.
- During Sheet Metal forming, wearing cotton hand gloves is compulsory.

General Rules:

- Follow the lab timing (2:00 pm – 5:00 pm). There will be two attendances: Initial attendance will be based on Weekly Quiz and final one consists of filling the job submission form. Those coming after 2.00 will mis the quiz and those coming after 2:15 pm will be sent back.
- Mobile phones are prohibited during the lab.
- Students must take permission of the Tutor, if they have to leave the lab before time, for any emergency.
- Project report must be submitted on the 6th turn. Any group not submitting the project report by this time will be appropriately penalized.
- There will be a Weekly lab quiz during first 5 experiments on questions related to the previous lab. On the 6th turn, there will be lab quiz first followed by lab exam.
- Every student should obtain a copy of the Manufacturing Processes Laboratory manual. You are required to bring your lab manual every day. Lab manual will also be available on the course website.

Suggested Reading Materials:

- **Preferred:** Fundamentals of Modern Manufacturing: Materials, Processes and Systems, Mikell P. Groover
- Fundamental of Manufacturing, G. K. Lal & S. K. Choudhury
- Manufacturing Engineering & Technology, S. Kalpakjian
- E. P. Degarmo: Materials & Processes in Manufacturing, Macmillan.

Material List:

The list of materials that students can assume to be available, is attached as Appendix-A. Parts must be designed using only these materials. Outside material/parts should not exceed 5%.

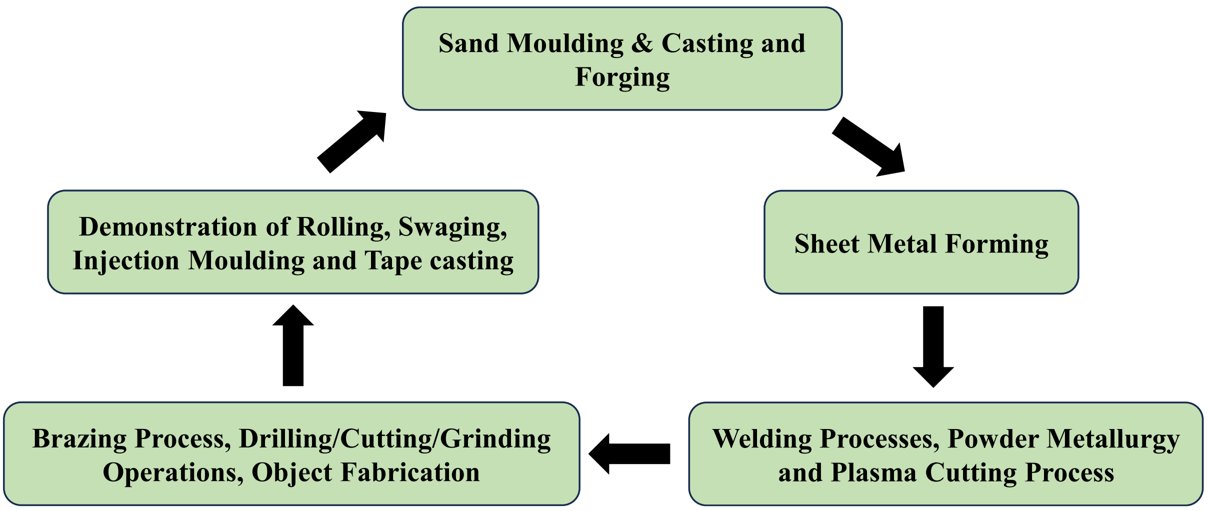
Plagiarism:

Plagiarised reports or drawings will be dealt with severely. You are encouraged to get inspiration and ideas from real things and also from internet. However, do not copy and present those things as your own.

Lab turns distribution:

S No.	Lab Turn	Experiments	Staff members
1	E1	Sand Moulding & Casting	Mr. Anurag Prasad (Assist Demo only), Raj Dipta, Surya Prakash
2	E2	Sheet Metal Forming	Mr. Rakesh Dixit/Anurag Prasad
3	E3	Welding Processes, Powder Metallurgy and Plasma Cutting Process	Mr. G. Mishra
4	E4	Brazing Process, Drilling/Cutting/Grinding Operations, Object Fabrication	Mr. B. R. Singh
5	E5	Demonstration of Rolling, Swaging, Injection Moulding and Tape Casting	Mr. Gyanendra Singh
6	LE6	**Lab Exam and Project Report Submission**	Respective staff member
7	P1	Project	Staff supervisor
8	P2	Project	Staff supervisor
9	P3	Project	Staff supervisor
10	P4	Project	Staff supervisor
11	P5	Project	Staff supervisor
12	P6	Project	Staff supervisor
13	PE	**Project Evaluation**	Respective members

Rotation of experimental turns:



There will be 5 Experiment groups consisting of approximately 15-20 students each. Each group will conduct different experiments on a given day (E1 – E5). In the next turn, they will be moved to the next experiment as per the rotation policy shown above.

Project Information

Schedule: Stick to the project schedule to earn marks for project progress.

1^{st} Turn	Project group formation
2^{nd} Turn	Bring a minimum of three project ideas along with the rough sketch (bring own drawings, NOT internet downloaded pictures)
3^{rd} Turn	Explore more details on the projects about how it can be fabricated in the lab, what process to be used and what materials are needed (bring more detailed drawings)
4^{th} Turn	Detailed discussion on a finalized project with proper drawing as per engineering norms, including parts drawing (with numbering and materials)
5^{th} Turn	A final discussion on drawing and process (bring draft of complete report)
6^{th} Turn	Final report submission (including drawings).
7^{th} - 12^{th} Turns	Project Fabrication (Bring hardcopy of your Project Drawings)
13^{th} Turn	Project Evaluation

Outline of project report: Keep a copy of the project report with you which you will use during your project turns.

S. No.	Description	Page no.
1	Project name	1
2	Tutor name	
3	Course In-charge and Lab In-charge names	
4	Section:	
5	S No. / Roll no. / Name / Signatures	
6	Contents	2
7	Certificate about plagiarism	3
8	Acknowledgements	4
9	Introduction	5
10	Motivation	6
11	Group member work distribution	7
12	Materials List	8
13	Isometric Drawing with numbering	9
14	Part Drawing (mm, page no.)	10 - xyz

Project report must include the following details:

- You must make isometric and part drawing with appropriate dimensions on A-4 sheet.
- Proper drawing should be made for each assembly and sub-parts, and submitted in consultation with tutor and lab staff.

- The manufacturing process involved in each component to be mentioned.
- The part drawing should be made with appropriate dimensions.

Project Constraints:

- Maximum size of the project: 40 cm x 40 cm x 40 cm (**to be followed strictly**) and total weight for casting objects should not exceed 1 kg of aluminium per project. Oversize/overweight projects will invite penalty in the project evaluation.
- Do not grind the aluminium parts of the project.
- Total project weight should not exceed 5 Kg.
- External colour /paint should not be used.
- One small 3-d printed part can be used in project

About the project

1. Plan your project carefully. Do not make it unnecessarily complicated. The project has to be entirely your work. Laboratory staff will be available only to guide you.
2. Your tutor, course In-charge and the technical staffs will advise on the projects.
3. There will be no extra lab turn for project.
4. The project groups will be formed and informed to you by the end of the <u>first lab turn</u>.
5. You should come with at least three ideas with the rough sketch on the <u>second lab turn</u> for the discussion. On the <u>third lab turn</u>, one idea will be frozen.
6. On the <u>fourth and fifth lab turns,</u> you should come with all necessary information, such as drawing, manufacturing process for each part etc. The drawing should be as per the engineering norms.
7. The copy of the final project drawing with material list and process plan (complete report) must be submitted on the <u>sixth lab turn</u>. You should select materials from the list only. (The list will be displayed on lab notice board).
8. The exact responsibilities of each group member should be specified.
9. Maximum size of the project: 40 cm x 40 cm x 40 cm (to be followed strictly) and total weight for casting objects should not exceed 1 kg of aluminium per project. Oversize/overweight projects will invite penalty in the project evaluation.
10. Do not grind the cast components of the project.
11. Moving parts in your project will result in extra credit during evaluation.
12. External colour/paint cannot be used.

Appendix-A: Materials List

(Materials made available to students for project)

Sr. No.	Items	Size	Cost
1	Mild Steel Flat	25x3 mm	Rs.65/kg
2	Mild Steel Flat	25x5 mm	Rs.63/kg
3	Mild Steel Round Rod	25 mm dia	Rs.65/kg
4	Mild Steel Round Rod	10 mm dia	Rs.68/kg
5	Mild Steel Round Rod	8 mm dia	Rs.68/kg
6	Mild Steel Round Rod	6 mm dia	Rs.70/kg
7	Mild Steel Round Rod	5 mm dia	Rs.70/kg
8	Mild Steel Round Rod	4 mm dia	Rs.70/kg
9	Mild Steel Round Rod	3 mm dia	Rs.80/kg
10	Mild Steel Square Rod	10x10 mm	Rs.63/kg
11	Mild Steel Square Rod	6x6 mm	Rs.65/kg
12	Mild Steel Round Pipe	25 mm dia (1" dia)	Rs 450/20 ft length pipe
13	Mild Steel Round Pipe	18 mm dia (3/4" dia)	Rs 425/20 ft length pipe
14	Mild Steel Round Pipe	10 mm dia	Rs 300/20 ft length pipe
15	Mild Steel Square Pipe	25x25 mm	Rs 450/20 ft length pipe
16	Mild Steel Square Pipe	15x15 mm	Rs 300/20 ft length pipe
17	Mild Steel Angle	25x25 mm	Rs 450/20 ft length pipe
18	Mild Steel Disks	20,30,40,50,60,70mm dia x8-10mm Thick	Avg ~ Rs 20-25/disc
19	Galvanized Iron Sheet	0.35 mm	Rs 756/sheet
20	Galvanized Iron Sheet	0.5 mm	Rs 820/sheet
21	Mild Steel Sheet	0.5 mm	Rs 945/sheet
22	Mild Steel Sheet	0.7 mm	Rs 1058/sheet
23	Mild Steel Sheet	1.0 mm	Rs 1278/sheet
24	Mild Steel Sheet	2.0 mm	Rs 3090/sheet
25	Thermocol	1/2 inch	Rs 30/sheet
26	Thermocol	1 inch	Rs 65/sheet
27	Thermocol	1.5 inch	Rs 110/sheet
28	Thermocol	2 inch	Rs 160/sheet
29	Thermocol	3 inch	Rs 220/sheet
30	Fevicol	small size tube	Rs 5/tube
31	Sandpaper for thermocol	No. 80	Rs 7/sheet
32	Small Nut-Bolt with Mild steel Washer	All sizes	Avg ~ Rs 10/piece
33	Thin Galvanized Wire	1 mm and 2 mm dia	Rs 20/ meter
34	Aluminium & Cast iron	For casting	Rs 470/ kg
35	Cast Iron for melting	Ingots	Rs. 75/ kg